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Saveetha Institute of Medical And Technical Sciences

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Food Rescue and Donation Distribution Optimization Using the Knapsack Problem

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Course code: CSA0620

Course: DAA

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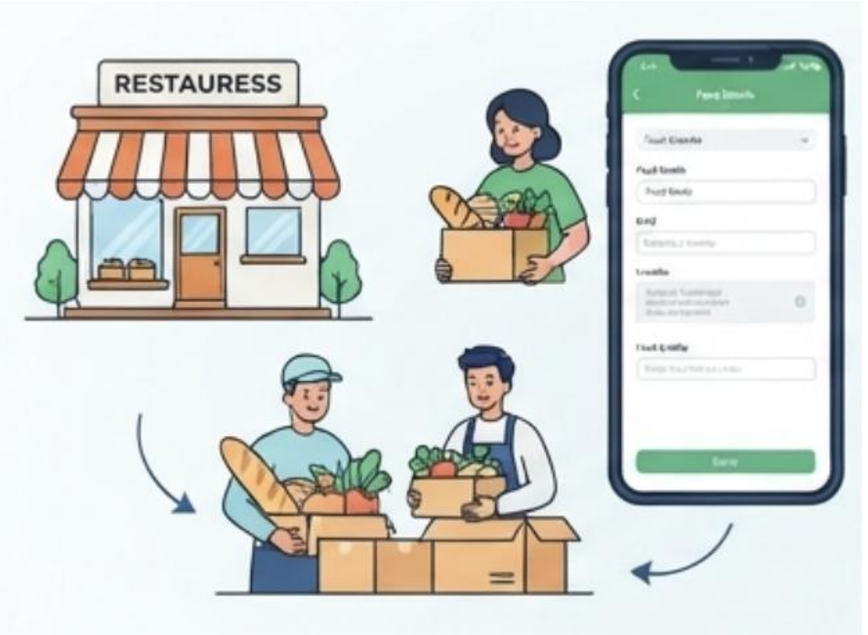
ABSTRACT

- Develops an app to connect food donors with NGOs/distribution centers.
- Donors can upload food type, quantity, weight, and expiry details.
- Uses the 0/1 Knapsack Algorithm to optimize food selection.
- Maximizes the useful food transported within limited vehicle capacity.
- Helps reduce food wastage and improve donation distribution efficiency.

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INTRODUCTION

- A food rescue app connects food donors with NGOs / food distribution centers.
- Donors can upload available surplus food.
- The app records food quantity, weight, expiry time and location.
- The system uses the Knapsack Algorithm to select the best food items based on available vehicle capacity.
- This helps reduce food wastage and improve donation distribution.



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Problem Statement

- Surplus food is available at different locations.
- Donation vehicles have limited carrying capacity.
- Manually selecting food may lead to wastage of space and resources.
- Some food may expire before distribution.
- A smart app is needed to optimize food collection and distribution.

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OBJECTIVES

- To develop an app for connecting food donors with NGOs and distribution centers.
- To collect and manage food donation details such as type, quantity, weight, and expiry time.
- To apply the 0/1 Knapsack Algorithm for optimal food selection.
- To maximize food distribution within the available vehicle capacity.
- To reduce food wastage and improve donation distribution efficiency.

Proposed System/App

- **Donor:** Uploads surplus food details.
- **App:** Stores donation information.
- **Optimization:** Knapsack selects the best combination.
- **NGO:** Receives the selected food.
- **Admin:** Monitors donations and distributions.

Example Scenario:

- A restaurant has 30 kg of surplus food.
- A donation vehicle can carry only 20 kg.
- The donor enters the food details in the app.
- The Knapsack Algorithm selects the best food items within the 20 kg capacity.
- The app displays the selected food items and NGO distribution details.



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Modules And Technologies Used

1. User Management Module
- Donor and NGO registration/login
- User profile management

2. Food Donation Management Module
- Add available food details
- Food name, quantity, expiry time, and location

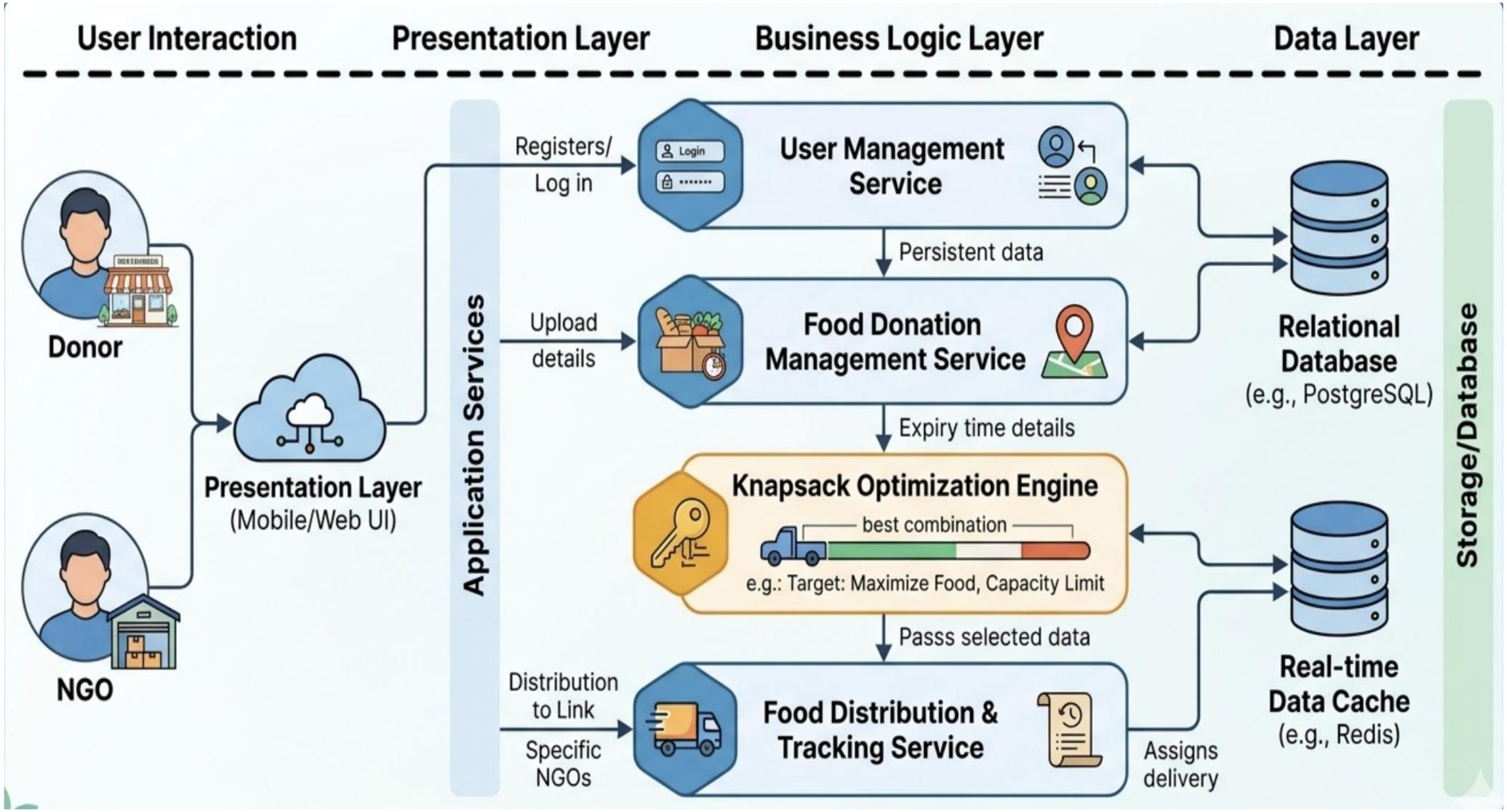
3. Knapsack Optimization Module
- Select the best combination of food donations
- Optimize food selection based on vehicle capacity
- Display selected food items and total quantity

4. Food Distribution & Tracking Module
- Assign selected food to NGOs
- Track donation and delivery status
- Maintain distribution history

- **Frontend:** HTML, CSS, JavaScript / React
- **Backend:** Python / Flask
- **Database:** MySQL
- **Algorithm:** 0/1 Knapsack
- **Optimization:** Dynamic Programming

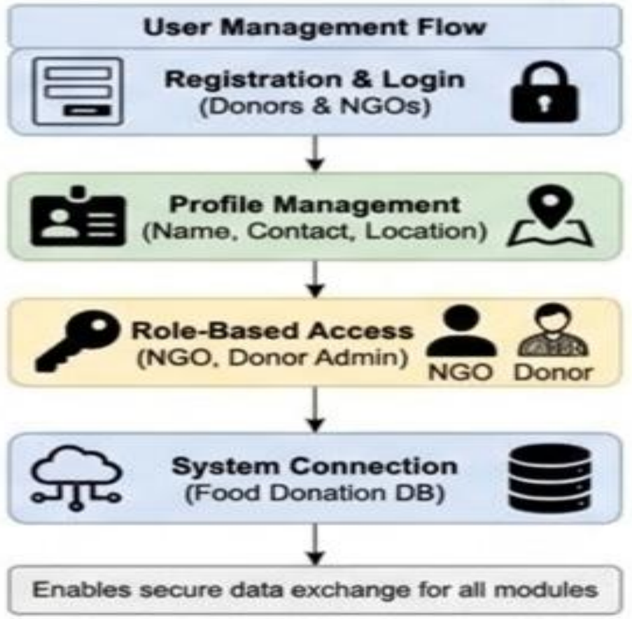
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System Architecture



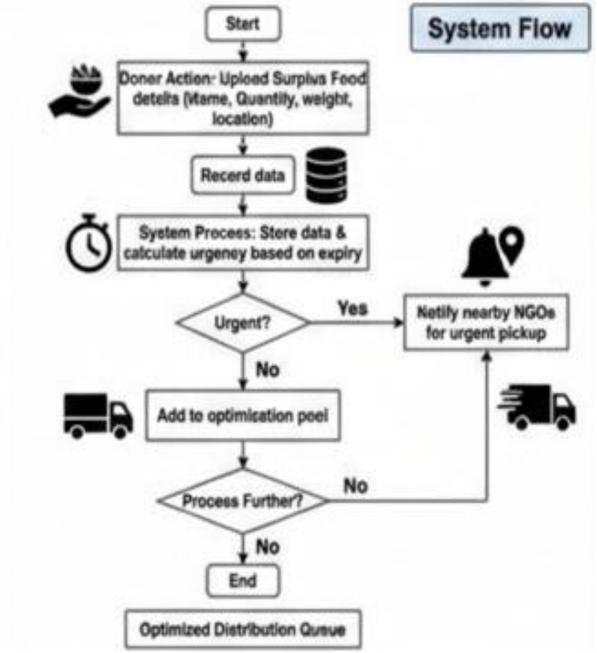
Module 1: User Management

- Provides registration and login functionality for donors and NGOs.
- Maintains user profile details such as name, contact and location.
- Provides role-based access for donors and NGOs.
- Allows users to securely manage their account information.
- Connects registered users with the food donation system.



Module 2: Food Donation Management

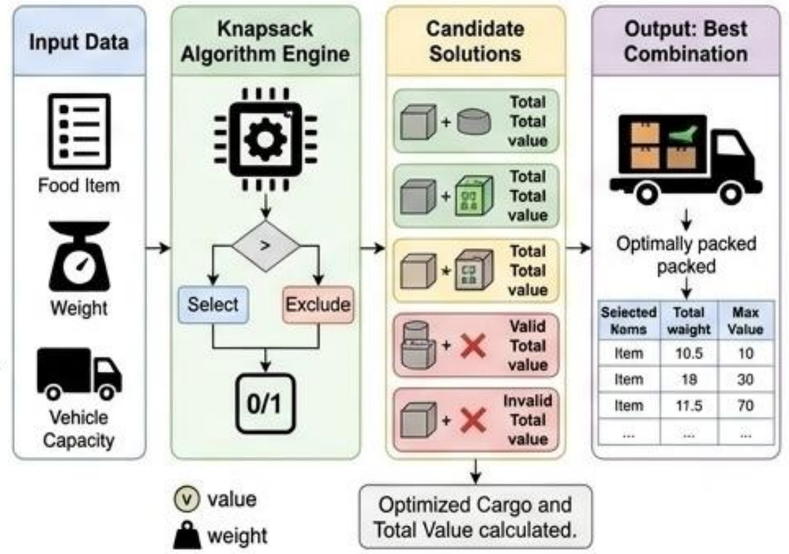
- Allows donors to add available surplus food to the application.
- Records food details such as name, quantity, weight and location.
- Stores expiry time to identify food that needs urgent distribution.
- Maintains donation records for further processing.
- Displays available food donations for optimization.



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Module 3: Knapsack Optimization

- Acts as the core algorithmic module of the proposed application.
- Takes food weight, value, expiry and vehicle capacity as input.
- Applies the 0/1 Knapsack Algorithm to find the optimal combination.
- Ensures the selected food does not exceed the available vehicle capacity.
- Displays the selected food items with total quantity and value.



Module 4: Food Distribution & Tracking

- Assigns the optimized food donations to suitable NGOs.
- Provides information about food collection and delivery.
- Tracks donation status from collection to final delivery.
- Updates the delivery status in the application.
- Maintains distribution history for future reference.



References

- H. Kellerer, U. Pferschy and D. Pisinger, Knapsack Problems, Springer, 2024.
- S. Martello and P. Toth, Knapsack Problems: Algorithms and Computer Implementations, John Wiley & Sons, 2025.
- D. Pisinger, “Where Are the Hard Knapsack Problems?” Computers & Operations Research, Vol. 32, No. 9, pp. 2271–2284, 2025.
- M. Goemans and D. P. Williamson, Approximation Algorithms, MIT Press, 2023.
- Food and Agriculture Organization (FAO), The State of Food and Agriculture: Moving Forward on Food Loss and Waste Reduction, FAO, 2024.


Thank
you!